BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 147 · Lesson 147 · Firewalls, Proxies and Encryption: Layered Defence

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What attack floods a service with requests to reduce availability?

Q2. What is a foreign key used for?

Q3. Which storage type is volatile?

Q4. Which layer in TCP/IP handles IP routing?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

firewall — Rule-based network traffic access control.

proxy server — Server that acts as intermediary for client requests.

TLS — Protocol protecting transport of application data.

defence in depth — Overlapping safeguards against different failures.

New learning

Filtering boundary — Firewall applies rules to traffic entering or leaving a network.

Mediating requests — Forward proxy can request resources on behalf of clients, cache and log access.

Protecting content — TLS encrypts data in transit; authentication and patching address other risks.

Worked example

A school wants to limit outgoing web browsing while protecting pupils' login details over public Wi-Fi. Select appropriate controls.

1. Client web request — Identify the facts and make an initial prediction.

2. Proxy and firewall controls — Explain the mechanism or trace the intermediate stage.

3. HTTPS encrypted transfer — A managed web proxy can filter/log permitted requests; a firewall can restrict traffic flows; HTTPS/TLS encrypts the connection so local eavesdroppers cannot read credentials. Policy, MFA and endpoint protection address further threats.

We Do · Practise with guidance

A small business has a firewall allowing HTTPS but an employee clicks a malicious attachment. Why was the firewall insufficient?

Traffic allowed on port 443

Identify the facts and make an initial prediction.

Malicious content reaches user

Explain the mechanism or trace the intermediate stage.

Endpoint controls matter

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

An allowed connection or email can carry malicious content. Endpoint malware detection, attachment scanning, patching, permissions and user training complement network filtering. A proxy may help inspect or block risky destinations.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State functions of firewall, forward proxy and TLS, one network example each. [6]

Question 2

Explain why encrypted traffic is not the same as authenticated trustworthy content. [6]

Question 3

Recommend a layered defence for a sixth-form network with remote learners, justifying four different controls. [8]

Exit Ticket · Answer before checking

A forward proxy and a firewall necessarily perform identical tasks.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A denial-of-service attack, potentially distributed (DDoS).

Q2. Referencing a candidate key in another relation.

Q3. RAM.

Q4. The internet layer.

You Do pages · Independent questions

Q1. Firewall controls traffic according to rule/state [2]; proxy intermediates requests/cache/filter [2]; TLS protects communication in transit [2].

Q2. Encryption protects data in transit [2]; a malicious website may still use encryption [2]; verify destination, credentials or content separately [2].

Q3. Credit four linked controls: firewall, proxy/web filtering, VPN or TLS, MFA, patching/endpoint protection, least privilege or monitoring [2 each].

Exit Ticket · Model answer

False. Proxy intermediates requests; a firewall applies traffic access controls.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: